SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

Department of Mathematics and Computer Science

SYLLABUS - UMAT-801 - Mathematical Modelling

L – T – P – C	0 – 0 – 4 – 4	**Prerequisites**	Calculus	Total Periods: **56**
Course Category	DSC			

Course Objectives

In this course, the students will be introduced to both deterministic and stochastic methods of mathematical modelling. The students will be introduced to effectively communicate the mathematics to achieve a realistic model of a given scenario.

Course Outcomes

- Able to understand the use of mathematics to solve a variety of problems across various disciplines.
- Able to appreciate the ideas and interconnections between various branches of mathematics.
- Able to apply core techniques of mathematical modeling to model, analyze and interpret real life scenarios.
- Able to formulate and communicate effectively with the mathematical models of situations.

Syllabus

Unit	Topic	Contents	Periods
1	**Deterministic analysis of observations**	Data transformations (linear models), model development (polynomial models), model evaluation (population modeling), the advantage of modeling (global warming modeling)	8
2	**Stochastic analysis of observations**	Model errors, optimal linear models	8
3	**Deterministic states**	Dimensionality analysis and similarity, applications of low complexity	8
4	**Stochastic states**	Probability density functions, models for probability density functions, data analysis, real distributions	8
5	**Case studies**	Case studies	24
		Total	**56**

Key Text(s)

- Stefan Heinz, Mathematical Modeling, Springer-Verlag Berlin Heidelberg, First Edition, 2011. [Chapters 1, 2.1–2.3, 3.1–3.3, 4]

Suggested/Additional Readings

- Fredric Y.M. Wan, Mathematical Models and Their Analysis, Harper and Row Publishers, New York.
- Frank R. Giordano, William P. Fox and Steven B. Horton, A First Course in Mathematical Modeling, Fifth Edition, Brooks/Cole CENGAGE Learning, 2014.
- Mike Mesterton-Gibbons, A Concrete Approach to Mathematical Modeling, John Wiley & Son.
- Michael Mesterton-Gibbons, A Concrete Approach to Mathematical Modeling, Addison-Wesley Publishing Company, 1989.
- Walter J. Meyer, Concepts of Mathematical Modeling, McGraw Hill Co., 1984.
- W.E. Boyce, Case Studies in Mathematical Modeling, Pitman Advanced Publishing Programs, 1981.